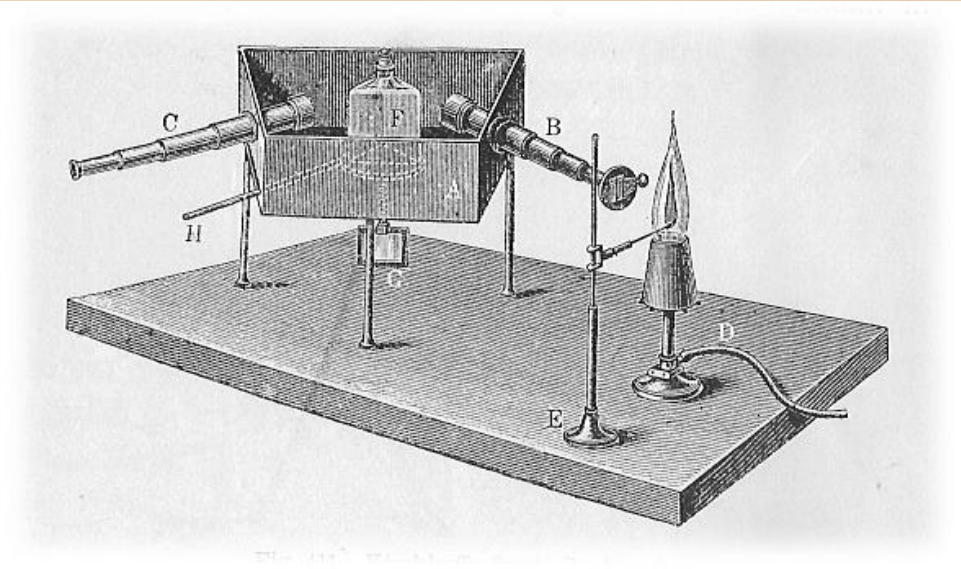
Modern Chemistry

Mendelev

Maria Curie



1850 – 1900

Outline

Earth, Water, Air and Fire was the basis of all things up until 18th Century

We have dealt with Basic Elements with Classical Chemistry

In 18th Lavasoir, Dalton, Berzelius began the scientific study of chemistry

Today we look the forces that bond elements at the atomic level

Elements

Dalton gave oxygen a weight of 7 compared to hydrogen being 1.

This is because Dalton selected the simplest ratios for compounds and concluded water was HO not H_2O as we know now.

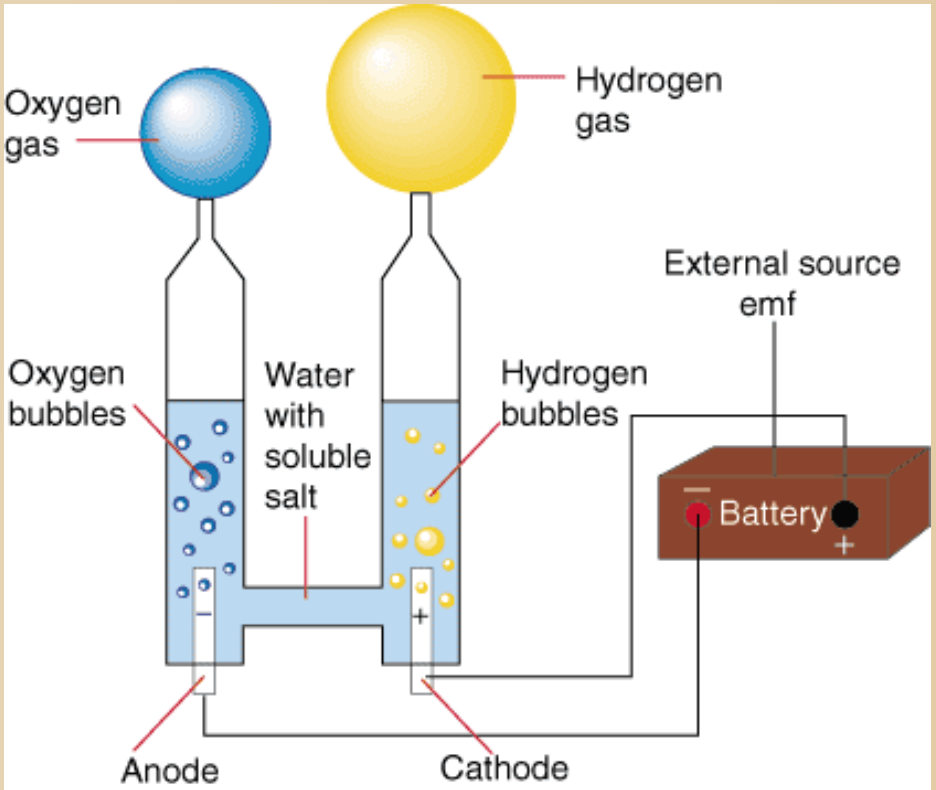
Modern data indicate that the combining ratio of hydrogen to oxygen atoms is 2:1 instead.

If the atomic weight of hydrogen is taken to be 1, that of oxygen must be 16 in compound.

Electrolysis - 1800

The invention of electric battery or pile by Volta allowed Davy and his protégé Faraday to discover new elements by electrolysis.

In 1800, Nicholson, an engineer, had electrolyzed water to produce hydrogen and oxygen which support the atomic hypothesis.



Stoichiometry

Berzelius (1779–1848) analyzed thousands of compounds to find accurate relative weights. He invented the modern notation of chemical compounds that results in the *chemical algebra* which is called stoichiometry.

Stoichiometry is used to determine the amount of products/reactants that are produced/needed in a given reaction.

_____ H_2 + _____ O_2 = _____H_2O

Electric Forces in Chemical Affinity

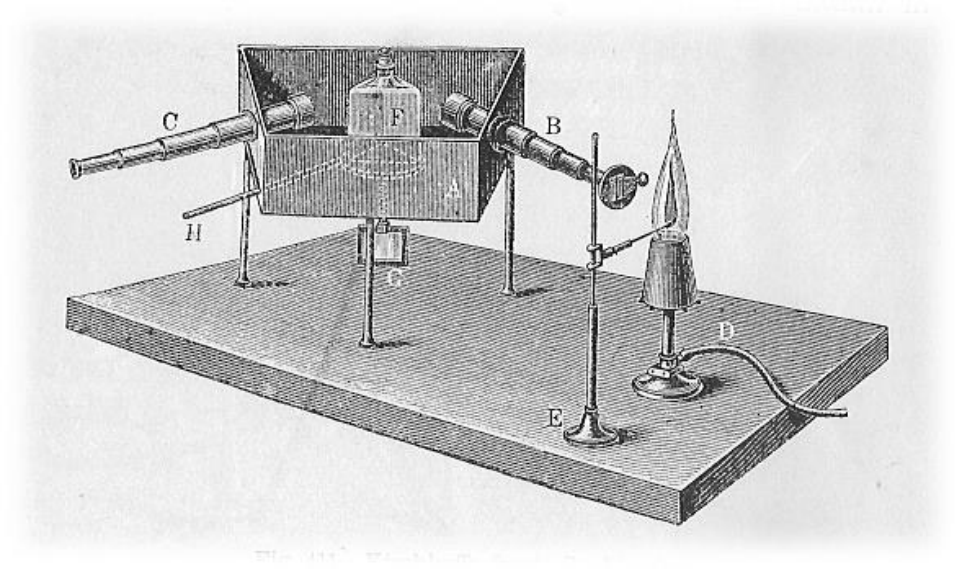
When salts like sodium chloride were melted and an electric current was passed through them, the metal always collected at the negative electrode (cathode) and the other element at the positive electrode (anode).

Berzelius concluded that electrical forces must be involved in chemical affinity. Berzelius thought that every atom had both a positive and a negative charge, one of which was greater than the other, and only oxygen was totally negative.

Spectroscope - Bunsen and Kirchoff

Various elements were known to give discrete spectra when heated.

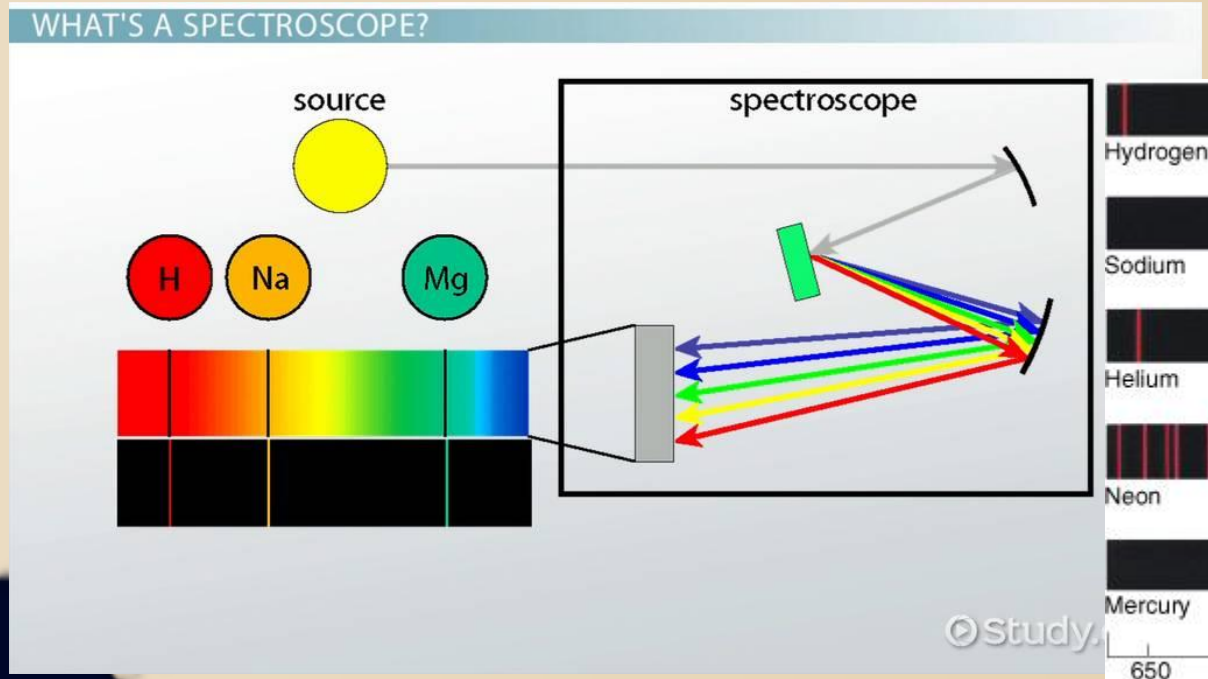
By dipping a platinum wire into a solution containing an element and then placing it in a flame, Bunsen and Kirchoff characterized the spectra of known elements.

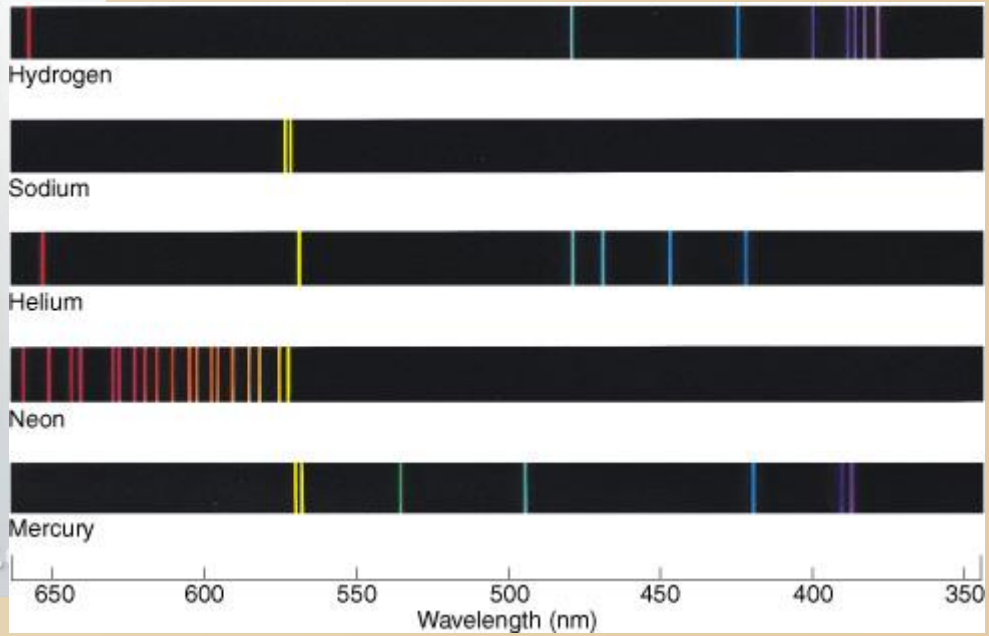


Spectroscope

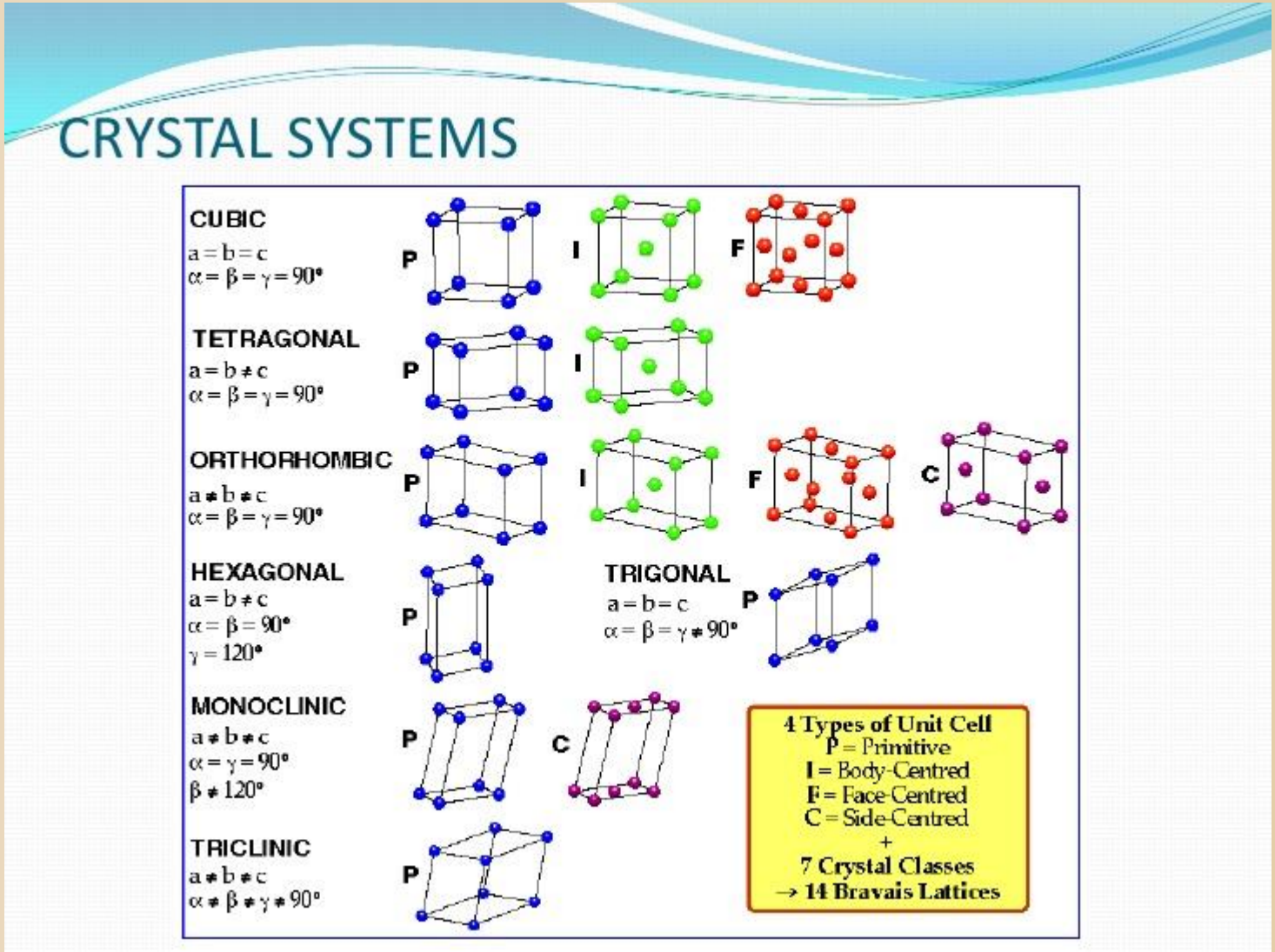
By placing the element in a fire, the light produced, when broken up by a prism, consisted of only a few spectra lines instead of the continuum of a rainbow.

Spectra lines were fingerprints of the elements due to crystalline





Crystalline Structure



Spectroscopy

Scientist looked at stars and could determine their elemental composition.

Later in sunlight the spectrum of hydrogen was seen along with a new spectrum of an element not yet found on Earth.

They realized the sun contained an unknown element and named it helium after the Greek Helios for sun.

The First International Conference – Chemistry, Karlsruhe, Germany in 1860

By 1860, there was chaos with the argument over what were the basic ratios of the atomic weights of elements.

Many reactions were known but arguments abounded as to which were the correct assignments.

The First International Chemistry Conference was held in Karlsruhe, Germany in 1860. Most of the leading chemists attended and Stanislao Cannizaro (1826–1910) proposed they reconsider the Avogadro hypothesis.

The First International Conference – Chemistry, Karlsruhe, Germany in 1860

By the end of the Conference all were in agreement and a table of atomic weights had been established. Using Avogadro's equal volumes assumption:

1 vol H + 1 vol Cl = 2 vol HCl

$$H_2 + Cl_2 = 2HCI$$

2 vol H + 1 vol O = 2 vol Water

$$2H_2 + O_2 = 2H_2O$$

Dmitri Mendeleev (1834–1907)

Confusion of thousands of compounds with no rhyme or reason.

Mendeleev started up the table of elements, in order of their atomic weights, every so often their properties started to repeat.

In 1869, he made a table of repeating periods. When an element didn’t seem to be similar in properties to the one above it, he skipped a position. These elements were unknown elements that would eventually be found.

Mendelev's Periodic Tablee

Tabelle II.

Reihen	Gruppe I. — R^2O	Gruppe II. — RO	Gruppe III. — R^2O^3	Gruppe IV. RH^4 RO^2	Gruppe V. RH^3 R^2O^5	Gruppe VI. RH^2 RO^3	Gruppe VII. RH R^2O^7	Gruppe VIII. — RO^4
1	H = 1							
2	Li = 7	Be = 9,4	B = 11	C = 12	N = 14	O = 16	F = 19	
3	Na = 23	Mg = 24	Al = 27,3	Si = 28	P = 31	S = 32	Cl = 35,5	
4	K = 39	Ca = 40	— = 44	Ti = 48	V = 51	Cr = 52	Mn = 55	Fe = 56, Co = 59, Ni = 59, Cu = 63.
5	(Cu = 63)	Zn = 65	— = 68	— = 72	As = 75	Se = 78	Br = 80	
6	Rb = 85	Sr = 87	?Yt = 88	Zr = 90	Nb = 94	Mo = 96	— = 100	Ru = 104, Rh = 104, Pd = 106, Ag = 108.
7	(Ag = 108)	Cd = 112	In = 113	Sn = 118	Sb = 122	Te = 125	J = 127	
8	Cs = 133	Ba = 137	?Di = 138	?Ce = 140	—	—	—	— — — —
9	(—)	—	—	—	—	—	—	
10	—	—	?Er = 178	?La = 180	Ta = 182	W = 184	—	Os = 195, Ir = 197, Pt = 198, Au = 199.
11	(Au = 199)	Hg = 200	Tl = 204	Pb = 207	Bi = 208	—	—	
12	—	—	—	Th = 231	—	U = 240	—	— — — —

Modern Periodic Table http://www.ptable.com/

Anions/Cations

Arrhenius found that pure water does not conduct an electric current and neither does a salt like sodium chloride (NaCl). But salt dissolved in water conducts electricity.

His theory was that some molecules dissolved in water forming a solution that had charged particles or ions.

$$NaCl \rightarrow Na^+ + Cl^-$$

Positive ions are called cations and negative ions, anions. It was these ions that conducted the current from one electrode to the other.

Arrhenius also proposed the first theory of acids and bases. Acids, according to him were compounds that produced H^+ in solution and bases produced OH^-

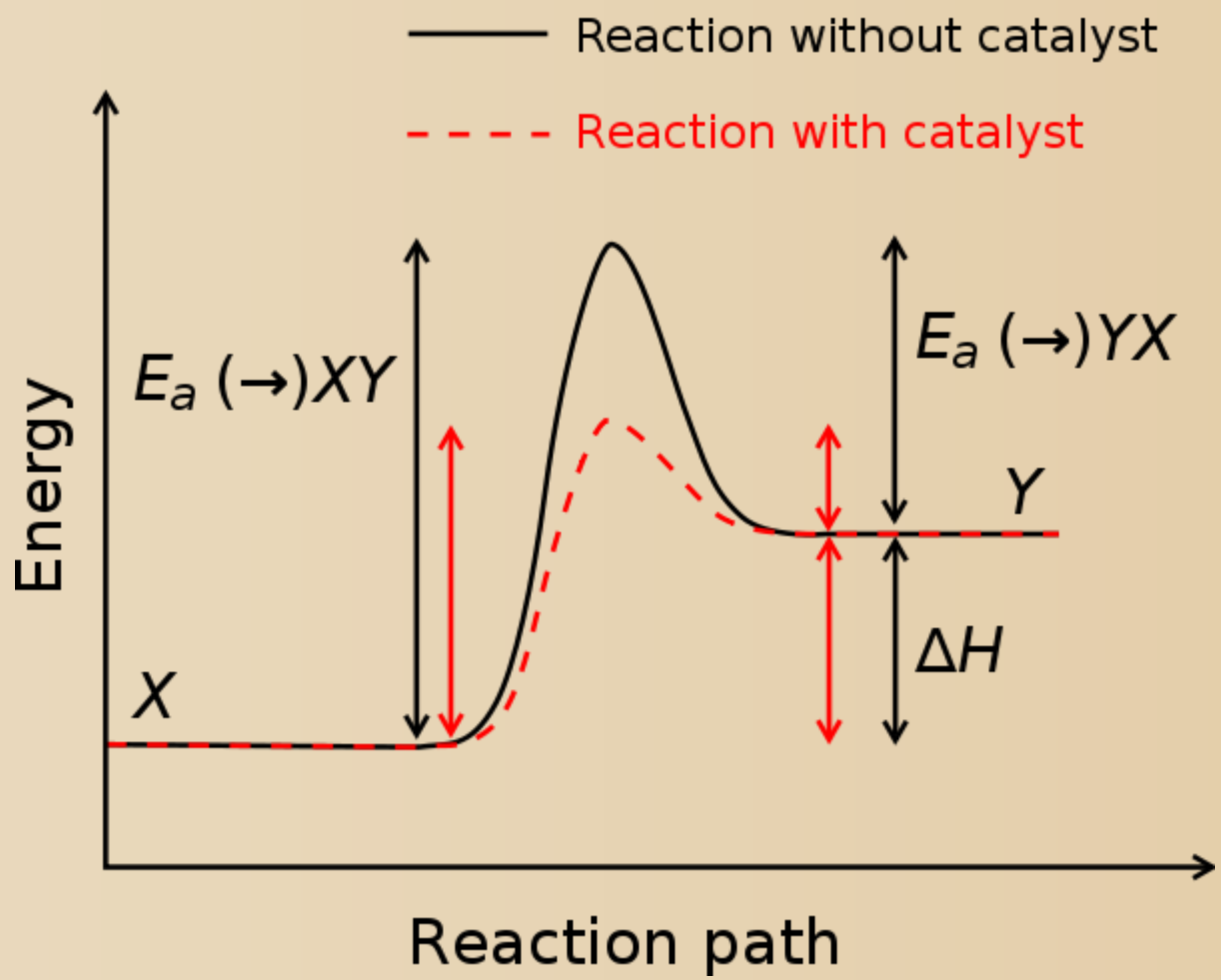
Activation Energy

Arrhenius gave us the concept of activation energy for a reaction around 1888.

Arrhenius determined that an energy barrier – the difference in initial energy of reactants and the necessary higher energy that must be reached at some point in the reaction – controls the rate of a reaction.

Activation Energy

Arrhenius actually gave a quantitative formula for determining activation energies.



Modern Chemical Bonding Theories

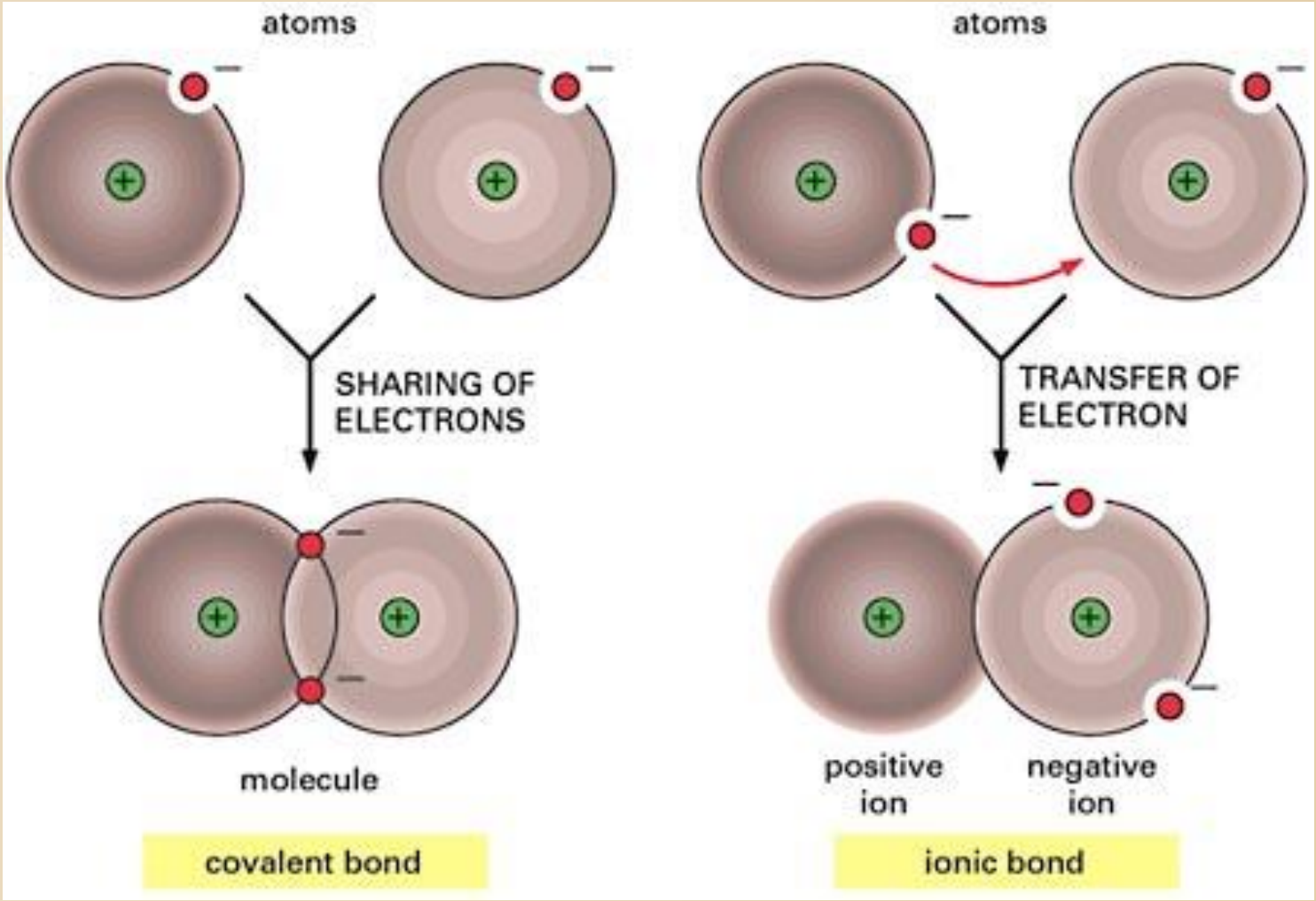
It would be in the 20 th century, with first the discovery of subatomic particles (particularly the electron) and later the development of quantum mechanics that chemists would have our modern theories of chemical bonding.

Two American chemists, who both spent most of their careers in California, G.N. Lewis and Linus Pauling would be the central figures in the development of bonding theory.

Pauling's The Nature of the Chemical Bond and the Structure of Molecules and Crystals (1939) is a comprehensive treatise on early theories of chemical bonding.

Chemical Bonds through Electrons

Chemical bonds through the sharing of atom (Covalent) or Transfer of Electron

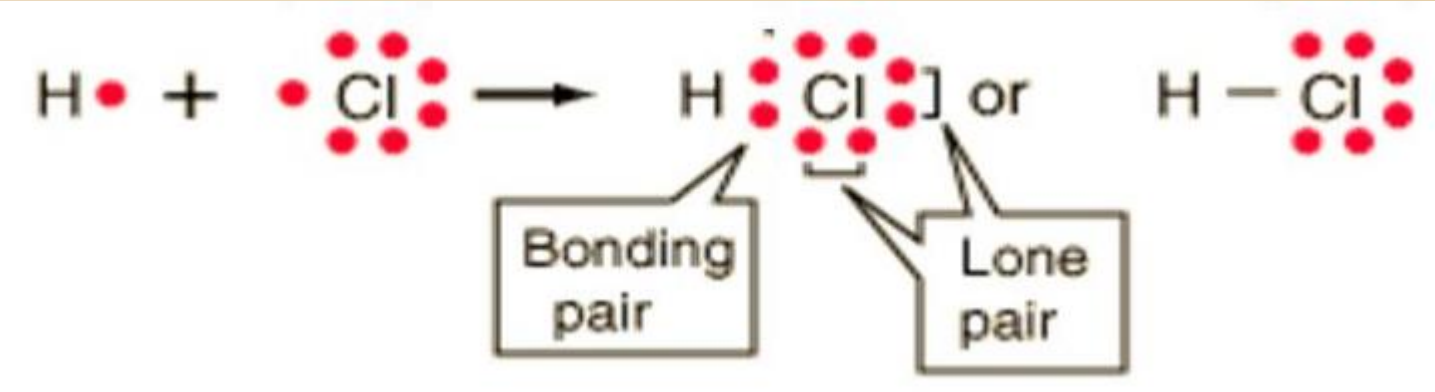


Chemical Bonds – Electric Forces

In Covalent Bonds Equal Sharing of Atoms cause No Polarity (Non-Polar) vs Unequal Sharing of them cause Polarity. (Polar)

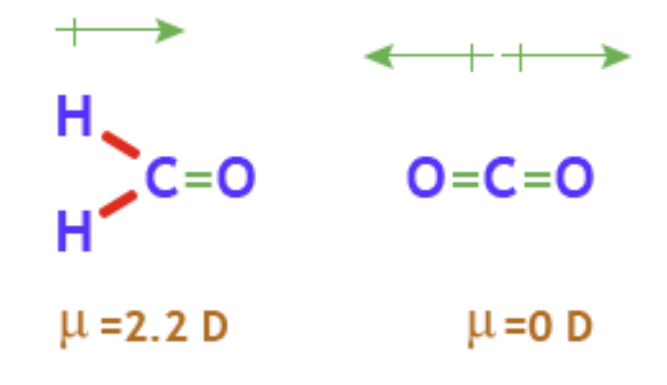
Non-Polar:

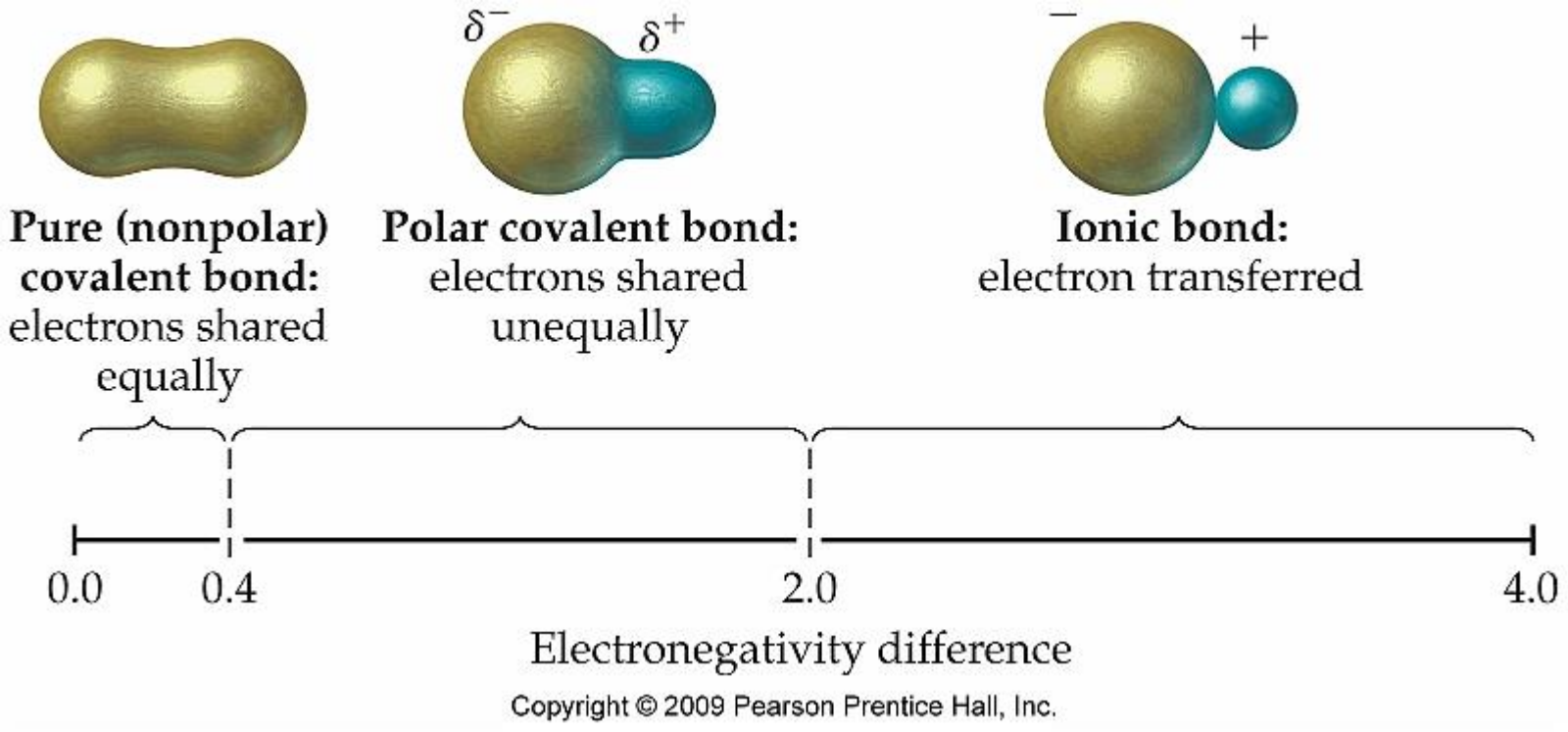
Polar:



Dipole Moment

A large difference in electronegativity between two bonded atoms will cause a dipole.





The Mystery of Matter

UNRULY ELEMENTS

Over a single weekend in 1869, a young Russian chemistry professor named Dmitri Mendeleev invents the Periodic Table, bringing order to the growing gaggle of elements.

But this sense of order is shattered when a Polish graduate student named Marie Sklodowska Curie discovers radioactivity, revealing that elements can change identities

— *and that atoms must have undiscovered parts inside them.*